IMPACT OF SCREEN TIME:

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Research Questions:

1.) Does productivity level differ by screen time bracket?
2.) Does attention span level differ by screen time bracket?
3.) How does screen time differ by education levels?
4.) Are certain age groups more likely to have a higher screen time?

Motivation:

I care about this specific issue, as it is an event that has rapidly increased in significance recently. Specifically, as society continues to technologically develop, and develop access to more online tools, screen time inevitably increases, whether it's for work, studying, or personal entertainment. However, screen time is also a major issue in the younger generation of this era, as I have personally noticed this. These research questions uncover where screen time is most common statistically, and its clear impact in multiple aspects. Knowing the answers to these affects our understanding of this development, by observing the possible positive and negative consequences, and predicting what future generations might be doing cognitively. Analyzing this can also facilitate research on if future devices can be modified to incorporate more positive features, rather than negative.

Dataset:

The dataset that I have chosen is from Kaggle and can be accessed through this link: https://www.kaggle.com/datasets/muhammadalirazazaidi/screen-time-data-productivity-and-attention-span?resource=download

Note: To download the dataset as a CSV file, one will have to sign into Kaggle first.

This dataset contains 15 descriptive columns including: age group, gender, education level, occupation, average daily screen time, device, screen activity, app category, screen time period, preferred work environment, productivity, attention span, work strategy, notification handling, and usage of products. For my research questions, I will not be utilizing all of these columns.

Overall, this dataset contains much data providing insight into the preferences and impact based on screen time, as well as the details about each respondent. This will overall allow for a deep analysis that will be effectively able to answer my research questions, as well as fulfill my motivation.

Challenge Goals:

Selected 2 Goals:

1.) Incorporating a new Python Library

I believe that this project will meet this goal the research questions developed for this project will require data visualization for effective analysis. By importing new types of data visualization libraries, this project will be to use more unique types of data visualizations to further enhance the effectiveness of the message conveyed. This project will meet this goal because it will further illustrate specific patterns about how screen time affects a variety of factors.

2.) Using Machine Learning

Part of this project is to analyze the specific decisions or preferences made based on both the details of a respondent, but also their screen time levels. With these variables, incorporating machine learning to predict possible scenarios using models will be effective on testing on how reliable, or how strong a trend is in this dataset. This project will meet this goal because it will further emphasize a specific trend, or pattern seen in this data.

Method:

RQ1: Does productivity level differ by screen time bracket?

1.) Group the dataset by screen time bracket, then use groupby to calculate the percentage of respondents falling into each productivity level within each bracket.

2.) For each screen time bracket, create a separate pie chart using the Plotly library, using the calculated percentages as the input values.

3.) Compare the pie charts across brackets to observe how the distribution of productivity levels shifts as screen time increases.

This will allow for a conclusion on the first research question, by visually analyzing how productivity is distributed as screen time increases, from pie chart to pie chart. This also uses the challenge goal by using an external library to create a unique data visualization.

RQ2: Does attention span level differ by screen time bracket?

1.) Group the dataset by screen time bracket.

2.) Within each group, count the number of responses falling into each attention span interval.

3.) Using the Plotly library, create a separate funnel chart for each screen time bracket, and compare the counts at each stage of the funnel to identify differences in attention span distribution across brackets.

This will allow for a conclusion on the second research question, by also visually analyzing how attention span levels change based on screen time. By creating separate funnels, it is visually simple to see which bars are smaller, or larger based on how many respondents fall into to that category. This also aims for the challenge goal, as it uses an external library to create a unique data visualization.

RQ3: How does screen time differ by education level?

1.) Group the dataset by education level.

2.) Within each group, count the number of respondents falling into each screen time bracket.

3.) Convert each screen time bracket into a numeric value (its midpoint), then calculate the average screen time for each education level by taking the mean.

4.) Using the Plotly library, create a bar chart with education level on the x-axis and average screen time (in hours) on the y-axis.

This will allow for a conclusion on the third research question, by seeing how long bars are for different screen time intervals, for different education levels. That will visually show which education levels contain the most, and the least screen time levels. Similar to the previous two research questions, this also uses an external library to develop a data visualization, following the challenge goal.

RQ4: Are certain age groups more likely to have a higher screen time?

1.) Convert each screen time bracket into a single number (its midpoint), so it can be used as the target the model tries to predict.

2.) Convert Age Group into a numeric format (ordinal encoding, since the age groups have a natural order) so the model can use it as an input.

3.) Split the data into a training set and a testing set, using Age Group as the input and screen time (the numeric version) as what the model is trying to predict.

4.) Train an XGBoost Regressor on the training set, then have it make predictions on the testing set.

5.) Compare the predicted screen time values across age groups to see if there's a clear trend and check how accurate the predictions are using mean absolute error.

This will allow for a conclusion for the fourth research question, by using a machine learning model to predict on data. These predictions will showcase a trend that can be used to highlight if certain age groups are more likely to have a higher screen time, and specifically which ones. This follows the challenge goal as it uses a machine learning model to analyze a specific trend for a conclusion.

Work Plan:

4 Tasks: Each one tackling each research question

Task 1: Creating multiple pie charts to determine how productivity is affected
Task 2: Creating multiple funnel visualizations to observe how attention span is affected
Task 3: Creating multiple bar charts for each education level
Task 4: Using Machine Learning to predict screen time based on a given age group

With debugging, testing, planning, and coding, each task will take approximately 2-4 hours depending on potential bugs.

I will be using VS Code as my environment to create this project. Referring to my workflow, for each task I will first create a step-by-step plan on how to create it. This will include the tools I will use, and what the output should look like. Then after developing and programming, I will ensure to test all edge cases, and check if my output matches the desired output, and if it effectively answers the research question, and links back to my motivation.